Advanced Statistics 2027-1

Probabilistic Machine Learning

Course number: TBD

Lecturer	Opher Donchin
Tutorials	None
Communication	By e-mail
Office hours	By appointment

Syllabus

An introduction to machine learning from a probabilistic and theoretically driven perspective. The course follows the approach of the undergraduate statistics course (367-1-4361 Statistical Computation and Analysis), placing probabilistic modeling and Bayesian uncertainty propagation at the center. The focus is on how methods relate to each other and to the underlying model of the data.Conceptual development is accompanied by practical work in which students fit models, examine their predictions, diagnose common failures, and interpret uncertainty.

Topics include probability and statistical foundations; graphical models and Bayesian decision theory; entropy, KL divergence, and mutual information; automatic differentiation and stochastic optimization; generative and discriminative classification; logistic regression; ridge and lasso regression; splines and generalized additive models; generalized linear models; multilayer perceptrons; backpropagation; neural-network training and regularization; Bayesian neural networks; convolutional neural networks; calibration; out-of-distribution detection; and selective prediction.

Practical work will use Python within Colab, principally NumPy, SciPy, scikit-learn, PyMC, ArviZ, and PyTorch. Each weekly meeting combines a two-hour lecture with approximately one hour of notebook-based guided or independent work.

Additional materials

1. Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022 (Book 1). Online edition.
2. Kevin P. Murphy, *Probabilistic Machine Learning: Advanced Topics*, MIT Press, 2023 (Book 2). Online edition.
3. Course slides, notebooks, exercises, project guidance, and supplementary readings will be distributed through the course site and the public course repository.

The schedule identifies the sections used to prepare and support each lecture; it does not imply that every derivation or example in those sections will be presented in class.

Prerequisites

There are no formal prerequisites. A background in mathematical probability, Bayesian statistics, linear algebra, regression, and Python programming will be helpful. Students who have not taken 367-1-4361 Statistical Computation and Analysis, or equivalent coursework, should consult the instructor before registering. Mathematical material will be refreshed and extended when it is needed in the course.

Course requirements and grading

The course grade is based on four components:

Component	Contribution
AI-based tutoring	10%
Notebook submission	10%
Project presentation	30%
Final exam	50%

Schedule

Lecture 1 — 21.10.2026

Probability and statistics review and extension

Topics Machine-learning tasks; probabilistic modeling; MLE, MAP and regularization; multivariate Gaussian models; exponential families; exchangeability; empirical Bayes and model averaging.
Book 1 Ch. 1; §§2.1–2.6, 3.1–3.3, 3.6, 4.1–4.3 and 4.5–4.7.
Book 2 Ch. 1; §§2.3–2.4, 3.2.5, 3.7 and 3.8.2.

Lecture 2 — 28.10.2026

Graphical models and Bayesian decision theory

Topics Factorization and conditional independence; plate notation; graphical models as probabilistic abstractions; loss, utility, classification and probabilistic prediction.
Book 1 §§5.1–5.1.6.
Book 2 §§4.1, 4.2.1, 4.2.4, 4.2.8 and selected material from §4.5.1.

Lecture 3 — 04.11.2026

Information theory for probabilistic prediction

Topics Entropy, cross-entropy, perplexity, KL divergence, forward and reverse KL, mutual information, sufficient statistics, and connections to likelihood and Bayesian inference.
Book 1 §§6.1.1–6.1.5, 6.2.1–6.2.2, 6.2.5–6.2.6, 6.3.1–6.3.4 and 6.3.8–6.3.9.
Book 2 §§2.7.1–2.7.2, 5.1.3, 5.1.6–5.1.8 and 5.3.3–5.3.4.

Lecture 4 — 11.11.2026

Mathematical preparation for optimization

Topics Eigendecomposition and SVD as geometric tools; gradients, Jacobians and Hessians; computation graphs; local, global, convex and nonconvex optimization.
Book 1 §§7.4.1–7.4.4, 7.5.1–7.5.3, 7.8 and 8.1.
Book 2 §§6.2–6.2.2.

Lecture 5 — 18.11.2026

Gradient-based model fitting

Topics First-order methods, learning rates, momentum, Newton's method as a conceptual contrast, stochastic gradient descent, automatic differentiation, and optimization of finite-sum objectives.
Book 1 §§8.2, 8.3.1 and 8.4.
Book 2 §§6.3.1–6.3.3.

Lecture 6 — 25.11.2026

Generative classification

Topics Gaussian discriminant analysis; linear and quadratic decision boundaries; naive Bayes; generative versus discriminative modeling; missing features; scoring and calibration.
Book 1 §§9.1–9.2.5 and 9.3–9.4.
Book 2 §§14.1–14.2.2.

Lecture 7 — 02.12.2026

Discriminative classification and Bayesian logistic regression

Topics Binary and multinomial logistic regression; MLE and MAP; standardization; class imbalance; priors; posterior approximation and posterior predictive probabilities.
Book 1 §§10.1–10.2.4, 10.2.7–10.2.8, 10.3.1–10.3.5 and 10.5.
Book 2 §§15.3.3–15.3.7.

Lecture 8 — 09.12.2026

Regularized linear models and generalized linear models

Topics Ridge and lasso; sparsity and regularization paths; splines and generalized additive models; robust regression; multicollinearity; Bayesian shrinkage and variable selection; binomial and Poisson GLMs.
Book 1 §§11.3, 11.4.1–11.4.5, 11.5–11.6, 11.7.6–11.7.7 and 12.1–12.3.
Book 2 §§15.2.5–15.2.8.

Lecture 9 — 16.12.2026

Multilayer perceptrons and backpropagation

Topics Nonlinear representations; activation functions; depth; forward and reverse differentiation; vector-Jacobian products; backpropagation and computation graphs.
Book 1 §§13.1–13.2.5 and 13.3.
Book 2 §§16.1, 16.2.1–16.2.2 and 16.3.1.

Lecture 10 — 23.12.2026

Neural-network training, regularization and predictive uncertainty

Topics Learning rates; gradient pathologies; initialization; residual connections; early stopping; weight decay; dropout; overparameterization; MC dropout and deep ensembles as practical uncertainty methods.
Book 1 §§13.4.1–13.4.5, 13.5.1–13.5.5 and 13.5.7.
Book 2 §§17.2.1, 17.3.1 and 17.3.9–17.3.10.

Lecture 11 — 30.12.2026

Convolutional neural networks

Topics Convolution, padding, stride and channels; pooling; normalization; residual and dropout layers; constructing and fitting a convolutional classifier.
Book 1 §§14.1 and 14.2.1–14.2.4.
Book 2 §§16.2.3–16.2.6 and 16.3.2.

Lecture 12 — 06.01.2027

CNN architectures, image classification and uncertainty

Topics The progression from LeNet and AlexNet to Inception, ResNet and DenseNet; practical image classification; out-of-distribution inputs; selective prediction and the limits of predictive confidence.
Book 1 §§14.3.1–14.3.5, with particular emphasis on §14.3.4.
Book 2 §§17.4.6 and 19.3.2–19.3.3.

Lecture 13 — 13.01.2027

Project presentations and course synthesis

Topics Student project presentations connecting probabilistic assumptions, computation, prediction, evaluation, uncertainty and scientific interpretation.
Book 1 No new sections; projects may draw on Chs. 1–14.
Book 2 No new sections; relevant previously assigned sections may be used.